

MEDICAL IMAGE CLASSIFICATION

A Research Foundation & Project Plan

ThyroVision AI

An Explainable, Attention-Driven Deep Learning Framework for Thyroid Ultrasound Classification and Early Risk Assessment

Prepared for: Research Internship — Medical Image Classification (ML)

Audience: First-time researcher with no prior ML or medical background

Purpose: Literature foundation + scoped, defensible project proposal to discuss with your mentor

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How to use this document

This is not a textbook you read front to back. Parts 0–2 bring you from zero to literate. Part 3 onward narrows to your actual project (ThyroVision). The single most useful sections for your first mentor meeting are Part 11 (the project plan) and Part 13 (questions to ask). Skim everything once, then go deep where your mentor steers you.

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Part 0 — How Research Internships Actually Work

You said you don't know how research is done. That's the most important thing to fix first, because it changes how you read everything else. Here is the mental model.

0.1 What your mentor actually expects from you

A research internship is not a course with a syllabus. Nobody will hand you the answer because nobody has it yet — that's what makes it research. Your job in the first few weeks is to become the person in the room who knows the existing literature best, then identify a small, specific gap you can attack. Concretely, your mentor (Dr. Gunjan) expects you to:

- Understand the problem well enough to explain it back in plain language.
- Map what has already been tried, what worked, and where the field is stuck.
- Propose one narrow, achievable contribution — not 'solve thyroid cancer', but something like 'test whether adding attention module X improves benign/malignant calls on dataset Y, and show with Grad-CAM that the model looks at the nodule and not at image artifacts.'
- Run experiments, log results honestly (including failures), and iterate.

0.2 The core loop of research

Literature review → find a gap → form a hypothesis → design an experiment → run it → analyse → repeat → write up. You are currently at step one. The professor's suggested title has quietly done you a huge favour: it has pre-narrowed the gap for you. 'Medical Image Classification' is an entire field with thousands of papers; 'explainable attention-driven thyroid ultrasound classification' is a focused, publishable intern project. Lean into that narrowing — do not re-broaden it.

0.3 How to read a paper (the 3-pass method)

You will not read papers linearly. Use the widely-taught three-pass approach:

1. Pass 1 (5 min): Title, abstract, figures, conclusion. Decide if it's even relevant. Most papers stop here.
2. Pass 2 (30 min): Read intro and method at a conceptual level. What dataset? What model? What did they compare against? What was the headline result?
3. Pass 3 (deep): Only for the handful of papers you will build on. Understand it well enough to re-implement or critique.

Beginner tip

For every relevant paper, write four lines in a notebook: (1) Problem, (2) Method, (3) Result/metric, (4) Limitation. After 15 papers you'll start seeing the same limitations repeat — those repeated limitations are where your contribution lives.

0.4 Tools to manage the literature

- Zotero (free) — save papers, auto-grab citations, organise by tag. Install the browser connector.
- Google Scholar — set up alerts for 'thyroid ultrasound deep learning'; use the 'Cited by' link to walk forward in time from a key paper.

- Connected Papers / Semantic Scholar — visualise which papers cite which; fast way to find the 'landmark' nodes in a subfield.
- A single living document (this one) — keep a running 'paper log' table and a 'open questions for mentor' list.

Part 1 — Field Foundations (Just Enough ML)

This part gives you the minimum conceptual vocabulary to not be lost. It is deliberately not exhaustive. Depth comes later, on the specific methods your project uses.

1.1 What is medical image classification?

Image classification means assigning a label to an image: 'cat' vs 'dog'. Medical image classification assigns a clinically meaningful label to a medical scan: 'benign' vs 'malignant', or a disease grade, or a risk category. The model learns this mapping from examples (images that doctors have already labelled).

1.2 Why medical image classification is harder than 'cats vs dogs'

This distinction matters for your mentor meeting — it shows you understand the domain, not just the algorithm.

Dimension	Natural images (ImageNet)	Medical images
Data volume	Millions of labelled images	Often hundreds to a few thousand — labelling needs expert clinicians
Label cost	Cheap (crowdsourced)	Expensive; needs radiologists, and they disagree with each other
Ground truth	Obvious to anyone	Often requires biopsy/pathology to confirm — and even that is imperfect
Class balance	Roughly balanced	Severely imbalanced (e.g. ~5–10% malignant)
Cost of error	Trivial	A missed cancer (false negative) can be fatal
Trust	Not required	Clinicians won't use a black box; you must explain why the model decided what it did
Variation	Lighting, angle	Different machines, settings, operators — 'domain shift' is brutal

Takeaway: the three things that dominate medical imaging research — small/imbalanced data, the need for explainability, and generalisation across machines — are exactly the three things baked into your project title (attention to focus on the right region, explainability, and risk assessment). That is not a coincidence; it's a well-chosen project.

1.3 The 7 concepts you must be able to define

Supervised learning — the model learns from labelled examples (image → known answer). This is what you'll do.

Train / validation / test split — you train on one chunk of data, tune on a second (validation), and report final numbers on a third (test) the model has never seen. Mixing these up is the #1 way beginners get fake-good results.

Overfitting — the model memorises the training images instead of learning the general pattern; it scores great on training data and poorly on new data. Small medical datasets make this a constant danger.

CNN (Convolutional Neural Network) — the workhorse for images. It scans the image with small learnable filters that detect edges, then textures, then shapes, then whole structures (like a nodule). You stack many layers; early layers see simple patterns, deep layers see complex ones.

Transfer learning — instead of training from scratch (needs millions of images), you take a network already trained on ImageNet and fine-tune it on your few thousand thyroid images. This is standard practice and almost certainly what you'll do.

Attention — a mechanism that lets the model learn which parts of the image matter most and weight them more heavily. 'Attention-driven' in your title means the model is encouraged to focus on the nodule region rather than surrounding tissue or text overlays. It also doubles as a built-in explainability signal.

Transformer / Vision Transformer (ViT) — a newer architecture built entirely around attention. Splits the image into patches and lets every patch 'look at' every other patch. Powerful but data-hungry, so on small medical sets it's often used pre-trained or hybridised with CNNs.

Part 2 — The Clinical Problem: Thyroid & Ultrasound

You can't build a good medical model without understanding the medicine. This part is what turns you from 'a coder who used a thyroid dataset' into 'a researcher who understands the thyroid problem'. Mentors notice the difference immediately.

2.1 The thyroid, nodules, and why this matters

- The thyroid is a butterfly-shaped gland in the neck that regulates metabolism via hormones.
- A 'nodule' is a lump in the thyroid. They are extremely common — high-resolution ultrasound detects them in roughly 60–70% of adults, and some studies report up to ~76%.
- The crucial fact: the vast majority (about 90–95%) are benign. Only roughly 5–10% are malignant (cancerous).
- So the clinical task is a needle-in-haystack triage: find the few dangerous nodules without over-investigating the many harmless ones.

2.2 The overdiagnosis problem (this is your project's 'why')

Because nodules are so common and ultrasound is so widely used, the world has an overdiagnosis problem: huge numbers of harmless nodules get biopsied and even surgically removed. Reported unnecessary-biopsy rates for the common risk-scoring systems sit around 40%. The cautionary tale researchers always cite: a national screening program in South Korea increased thyroid-cancer diagnoses roughly 15-fold without changing the death rate — i.e. it found cancers that never needed finding, and people got surgery they didn't need.

Frame your project around this

Don't pitch ThyroVision as 'detect thyroid cancer'. Pitch it as 'reduce unnecessary biopsies while not missing real cancers' — i.e. improve the benign-vs-malignant triage and explain its reasoning so clinicians trust it. That framing is more clinically honest, more novel, and easier to defend. Sensitivity (catching cancers) and specificity (not over-flagging) become your real targets, not raw accuracy.

2.3 How ultrasound images are made — and why ML finds them hard

Ultrasound sends high-frequency sound pulses into the body; tissues reflect echoes, and the machine reconstructs a greyscale (B-mode) image from echo timing and strength. It is cheap, real-time, radiation-free, and the first-line tool for the thyroid. But for a machine-learning model it is one of the hardest modalities:

- Speckle noise — a grainy texture inherent to ultrasound physics; it's not random noise you can simply remove, and it can hide or mimic features.
- Operator dependence — the image depends on who held the probe, at what angle, with what gain/depth settings. Two scans of the same nodule can look different.
- Low contrast & unclear boundaries — nodule edges blur into surrounding tissue, making both segmentation and classification hard.
- Machine/vendor variation — different ultrasound machines produce different-looking images. A model trained on one hospital's machine often fails on another's. This is 'domain shift', the single biggest barrier to clinical deployment.

- Annotation artifacts — calipers, arrows, and text are often burned into the image; a careless model can 'cheat' by learning that 'images with a measurement caliper are malignant'. This is a classic data-leakage trap.

2.4 TI-RADS — the clinical scoring system you must know

TI-RADS (Thyroid Imaging Reporting and Data System), most commonly the American College of Radiology version (ACR TI-RADS), is the standardised way radiologists score a nodule's malignancy risk from its ultrasound appearance. The radiologist awards points across five feature categories:

- Composition (cystic/solid)
- Echogenicity (how bright/dark vs surrounding tissue)
- Shape (wider-than-tall vs taller-than-wide — taller-than-wide is suspicious)
- Margin (smooth vs irregular/lobulated)
- Echogenic foci (e.g. microcalcifications — suspicious)

Points sum to a category from TR1 (benign) to TR5 (highly suspicious), which — combined with nodule size — drives the decision to biopsy (fine-needle aspiration) or just monitor.

Category	Meaning	Approx. malignancy risk	Typical action
TR1	Benign	~0%	No FNA
TR2	Not suspicious	<2%	No FNA
TR3	Mildly suspicious	~2–5%	FNA if ≥2.5 cm; else follow
TR4	Moderately suspicious	~5–20%	FNA if ≥1.5 cm
TR5	Highly suspicious	>20%	FNA if ≥1.0 cm

Why this matters for your model design

TI-RADS is both your label source and your competition. Two ways to use it: (a) Benign-vs-malignant — collapse categories into a binary label (TR1–3 ≈ benign, TR4–5 ≈ suspicious) — the most common, cleanest task. (b) Predict the TI-RADS category itself (multi-class). A subtle but important caveat researchers raise: if your labels come from TI-RADS rather than biopsy-confirmed pathology, your model is learning to imitate a radiologist's scoring, not ground-truth cancer. State clearly which label source your dataset uses — mentors will ask.

2.5 The full diagnostic pathway (so you know where AI fits)

Ultrasound scan → TI-RADS scoring → (if indicated) Fine-Needle Aspiration biopsy → Bethesda cytology category → management. AI can assist at the TI-RADS step (standardise and de-bias the visual read) and at the triage step (reduce unnecessary FNAs). ThyroVision lives at the ultrasound-to-triage stage — before the needle.

Part 3 — Problem Formulation for ThyroVision

'Classification' hides several different tasks. Pick deliberately; each implies a different label, dataset, and metric.

Formulation	What the model outputs	Fit for ThyroVision?
Binary classification	Benign vs malignant (or 'biopsy / no-biopsy')	Best starting point — cleanest, most datasets support it
Multi-class (TI-RADS)	TR1–TR5 category	Good stretch goal; harder, labels noisier
Multi-label	Several co-occurring features (calcification + irregular margin...)	Advanced; needs feature-level annotations
Risk / probability	A calibrated malignancy probability + risk band	This is your 'risk assessment' component — layer on top of binary
Segmentation (support task)	Pixel mask of the nodule	Not classification, but very useful as a pre-step to crop the nodule

3.1 What the title's three pillars actually demand

Attention-driven → incorporate an attention mechanism (e.g. CBAM, SE-block, or self-attention/ViT) so the model focuses on the nodule. You will compare a baseline CNN vs the same CNN + attention to show the attention helped.

Explainable → produce visual explanations (Grad-CAM heatmaps, attention maps) showing the model looked at the nodule, not at artifacts. Optionally quantify trust. This is a first-class deliverable, not a nice-to-have.

Risk assessment / early diagnosis → output a calibrated probability and map it to a risk band, ideally aligned with TI-RADS so clinicians can interpret it. 'Early' = catching suspicious nodules at the triage stage.

Honest scoping note

All three pillars in one polished system is ambitious for a single internship through mid-July. The defensible MVP is: binary benign/malignant + one attention module + Grad-CAM explainability, evaluated with sensitivity/specificity/AUC and cross-dataset generalisation. Risk-band calibration is the natural 'if time permits' extension. Part 11 lays this out as phases.

Part 4 — Methods Landscape

4.1 Traditional ML & radiomics (know it, probably won't lead with it)

Before deep learning, the pipeline was: hand-design features (texture via GLCM, shape descriptors, intensity statistics — collectively 'radiomics'), maybe reduce dimensions with PCA, then feed an SVM, Random Forest, or XGBoost classifier. Strengths: works on tiny datasets, interpretable features, cheap to train. Weaknesses: features are hand-crafted and may miss subtle patterns; performance plateaus below deep learning when data is sufficient. Useful as a baseline to beat, and radiomics features can be fused with deep features in hybrid models.

4.2 CNN architectures — the menu

These are pre-trained backbones you fine-tune. You don't need to memorise internals; you need to know what each buys you and which to try first.

Architecture	Key idea	Trade-off for thyroid US
ResNet (18/50)	Skip connections let very deep nets train without vanishing gradients	The reliable default. Start here. ResNet-50 is the most common baseline in thyroid papers
DenseNet	Every layer connected to every later layer; reuses features	Strong with limited data; popular in medical imaging
Inception	Parallel filters of multiple sizes	Good multi-scale; heavier, older
EfficientNet	Balances depth/width/resolution via 'compound scaling'	Excellent accuracy-per-compute; great for limited GPU
MobileNet	Lightweight, depthwise-separable convs	For deployment/edge; lower ceiling but fast
ConvNeXt	Modernised CNN borrowing transformer design tricks	Strong recent baseline; needs a bit more data

4.3 Vision Transformers & hybrids

- ViT (Vision Transformer): pure-attention; powerful but data-hungry — usually needs pre-training to work on small thyroid sets.
- Swin Transformer: a hierarchical ViT with local windows; more efficient and a strong medical-imaging performer.
- Hybrid CNN-Transformer (e.g. TransUNet for the segmentation step): CNN captures local texture, transformer captures global context. Recent thyroid work increasingly uses these.

Practical recommendation

For an intern timeline on a small dataset: start with a fine-tuned ResNet-50 or EfficientNet baseline, then add an attention module and compare. Only reach for a full ViT/Swin if you have enough data (or use a pre-trained medical foundation model — see Part 7). 'Attention-driven' does not require a full transformer; an attention block on a CNN satisfies the title and is far more tractable.

4.4 Attention mechanisms (the heart of your title)

SE (Squeeze-and-Excitation) block — learns which feature channels matter and reweights them. Cheap, easy to drop into any CNN.

CBAM (Convolutional Block Attention Module) — adds both channel and spatial attention; the spatial part literally learns 'where' to look — ideal for focusing on the nodule. A very common, defensible choice for an 'attention-driven' framework.

Self-attention / transformer blocks — model relationships across the whole image; more powerful, heavier, more data-hungry.

Your experiment writes itself: Baseline CNN vs CNN + CBAM (or SE). If attention improves sensitivity/AUC and the attention maps land on the nodule, you have a clean, presentable result.

4.5 Explainability methods (the other half of your title)

Method	What it does	Use in ThyroVision
CAM / Grad-CAM	Heatmap of image regions that drove the prediction	Primary tool — show the model looks at the nodule. Easy, standard, expected
Grad-CAM++ / Score-CAM	Refined heatmaps, better localisation	Optional upgrade if Grad-CAM is too coarse
Attention maps	Visualise the attention module's weights directly	Free explainability if you use attention — pairs perfectly with your design
SHAP / LIME	Attribute importance to inputs/features	More natural for tabular/radiomic features than raw images; use if you fuse clinical data

A subtle research point worth raising with your mentor

Recent papers question whether Grad-CAM is genuinely 'faithful' — i.e. whether the heatmap truly reflects the model's reasoning. A small but real contribution for you would be to not just show heatmaps, but to sanity-check them (e.g. does performance drop appropriately when the highlighted region is masked?). That kind of critical rigour is exactly what separates a research intern from a tutorial-follower.

Part 5 — Data: Datasets & Pipeline

Your model is only as good as your data, and in thyroid ultrasound the public data is limited — which is itself a research opportunity. Below are the datasets actually used in the literature.

5.1 Public thyroid ultrasound datasets

Dataset	Approx. size	Labels	Notes / access
DDTI (TDID)	~480 images / ~390 patients	TI-RADS + segmentation	The classic open benchmark (Univ. Nacional de Colombia). Small, single-centre. Good for prototyping; weak for generalisation claims
TN3K	3,493 images / 2,421 patients	Benign/malignant + pixel masks	Multi-centre, multi-device (Zhujiang Hospital). Much more robust than DDTI. Widely used
TN5000	5,000 images	Benign/malignant + biopsy-confirmed	Newer (2025), open-access, biopsy confirmation (stronger ground truth). Comes with baseline benchmarks
TNSCUI / TN-SCUI 2020	~3,600+ images	Benign/malignant + segmentation	MICCAI 2020 challenge dataset; common benchmark
TG3K	~3,500 images (3,585)	Segmentation	Often combined with DDTI for segmentation training
TNUI	~1,381 images	Segmentation	Frequently held out as an external test set for generalisation
Stanford Thyroid US Cine-clip	Cine clips, ~167 patients	Radiologist + pathology	Real cine (video) data; richer, harder; via Stanford AIMI

Dataset strategy for a credible result

Train on a larger set (TN3K and/or TN5000), then test on a different dataset you never trained on (e.g. DDTI or TNUI) to demonstrate cross-dataset generalisation. Reporting only same-dataset numbers is the most common weakness reviewers attack. If you can show your attention model generalises better across datasets than the baseline, that alone is a paper-worthy finding.

Note on Indian data: there is no large, well-known public Indian thyroid-ultrasound dataset. If your NIT Delhi lab has any clinical collaboration to access even a small local set for external validation, that is a genuine novelty angle (domain shift + an under-represented population). Worth asking your mentor directly.

5.2 The data pipeline (and the traps)

DICOM → **image** — clinical scans come as DICOM files (pixel data + metadata). You'll convert to PNG/array, and must strip patient-identifying metadata (privacy).

Preprocessing — crop to the region of interest (ideally the nodule, using a segmentation step or provided masks), resize to a fixed size (e.g. 224×224), normalise pixel intensities.

Remove burned-in artifacts — calipers/arrows/text can leak the label. Crop or inpaint them out, or your 'great' accuracy may be the model cheating.

Augmentation — rotations, flips, small zoom/brightness shifts to artificially expand a small dataset and reduce overfitting. Be careful: don't flip in ways that destroy clinical meaning (e.g. taller-than-wide is a real signal).

Class imbalance — with ~5–10% malignant, use class weighting, oversampling the minority class, or focal loss. Otherwise the model learns to call everything benign and looks '90% accurate' while missing every cancer.

Splitting — by patient, not by image — if the same patient's images appear in both train and test, you've leaked information and your numbers are inflated. Split at the patient level. This is the single most common silent bug in medical-imaging student projects.

Cross-validation — with small data, use k-fold (e.g. 5-fold) cross-validation and report mean $\pm$ standard deviation, not a single lucky run.

Part 6 — Evaluation Metrics

Accuracy alone is misleading on imbalanced medical data. Here are the metrics that matter and what they mean clinically. (TP/TN = true positive/negative; FP/FN = false positive/negative, where 'positive' = malignant.)

Metric	Formula	What it tells you (clinically)
Accuracy	$(TP+TN)/(all)$	Overall correctness. Misleading when classes are imbalanced — avoid leading with it
Sensitivity / Recall	$TP/(TP+FN)$	Of the real cancers, how many did you catch? The most important metric — a missed cancer is the worst error
Specificity	$TN/(TN+FP)$	Of the benign nodules, how many did you correctly clear? Drives reduction of unnecessary biopsies
Precision (PPV)	$TP/(TP+FP)$	When the model says malignant, how often is it right?
F1 score	$2 \cdot (Prec \cdot Rec) / (Prec + Rec)$	Balance of precision and recall; useful single number under imbalance
ROC-AUC	Area under TPR-vs-FPR curve	Threshold-independent discrimination. The headline number in most thyroid papers (good models ~0.85–0.95)
PR-AUC	Area under precision-recall curve	More informative than ROC-AUC when the positive class is rare
Calibration (ECE / reliability)	Predicted prob. vs actual freq.	Are the probabilities trustworthy? Essential for your 'risk assessment' output

How to report results like a researcher

Always report sensitivity AND specificity AND AUC together (one alone hides failures), with confidence intervals or cross-validation mean $\pm$ SD. Show a confusion matrix. Compare against a baseline and ideally against TI-RADS/radiologist performance where available. Include the cross-dataset test. Numbers without a baseline comparison are not a result.

Part 7 — State of the Art (2023–Present)

Where the field is heading. You don't need to use all of this, but you must be able to discuss it — mentors gauge seriousness by whether you know the current frontier.

7.1 Foundation models & self-supervised learning

- The big shift: instead of training small models per task, the field is building large 'foundation models' pre-trained on huge unlabelled image collections via self-supervised learning (SSL), then fine-tuned on small labelled tasks. This directly addresses medical imaging's small-data problem.
- MedSAM (Nature Communications, 2024) — 'Segment Anything' adapted to medical images; a general segmentation foundation model.
- BiomedCLIP — a vision-language model pre-trained on biomedical image-text pairs; enables text-driven and zero-shot tasks (basis for MedCLIP-SAM / v2).
- Ultrasound-specific foundation models — e.g. UltraSAM and other 2024–2025 efforts building general ultrasound models; DINOv2-style SSL and masked-autoencoder approaches (e.g. USF-MAE) are being applied to ultrasound.
- Analogues that prove the pattern — RETFound (retina), EchoCLIP (echocardiography) show foundation models working per-modality; a thyroid/ultrasound equivalent is an active frontier.

7.2 Recent thyroid-specific highlights

- Multimodal + attention models — e.g. LLNM-Net (2025) fuses ultrasound images with radiology reports and demographics using bidirectional attention to predict lymph-node metastasis, reportedly outperforming human experts on that task. Signals the move toward multimodal thyroid AI.
- Interpretable attention networks — multi-attribute attention models that classify nodules and expose which TI-RADS-like features drove the call (interpretability built into the architecture, not bolted on).
- Multicentre clinical validation — 2024 studies (e.g. Ha et al.) test AI thyroid tools across multiple hospitals, the necessary step toward real deployment.

Where ThyroVision realistically sits

You are not going to out-build a hospital consortium's foundation model in one internship. Your edge is focus and rigour: a clean, reproducible, explainable attention model with honest cross-dataset evaluation — optionally leveraging a pre-trained backbone/foundation model as your feature extractor. 'We fine-tuned a strong pre-trained model, added attention, and rigorously showed it generalises and explains itself' is a perfectly respectable intern contribution.

Part 8 — Annotated Reading Map (the papers that matter)

The ChatGPT prompt asked for 50 paper summaries. That's the wrong tool for a beginner — you'd skim 50 and absorb none. Below is a curated, ordered map: read them roughly top to bottom. Use Google Scholar's 'Cited by' on each to expand. Build your own four-line log (Part 0.3) as you go.

8.1 Foundational ML/DL (understand the building blocks)

- AlexNet (Krizhevsky et al., 2012) — the paper that started the deep-learning era in vision. Read for context.
- ResNet (He et al., 2015) — skip connections; the backbone you'll likely use.
- DenseNet (Huang et al., 2017) and EfficientNet (Tan & Le, 2019) — efficient backbones for limited data.
- Attention Is All You Need (Vaswani et al., 2017) and ViT (Dosovitskiy et al., 2020) — the transformer/attention foundation.
- SE-Net (Hu et al., 2018) and CBAM (Woo et al., 2018) — the attention modules you'll actually add.
- Grad-CAM (Selvaraju et al., 2017) — the explainability method you'll use.
- U-Net (Ronneberger et al., 2015) — the segmentation workhorse, for your nodule-cropping pre-step.

8.2 Medical imaging & thyroid ultrasound (the core of your review)

- A survey on deep learning in medical image analysis (Litjens et al., 2017) — the classic field overview; read for the landscape.
- CheXNet (Rajpurkar et al., 2017) — landmark 'CNN matches radiologists' result (chest X-ray); useful template for how to frame a clinical DL paper.
- DDTI dataset paper (Pedraza et al., 2015) — your benchmark's origin.
- TN3K / thyroid-region attention segmentation (Gong et al., 2021) — key dataset + an attention-for-thyroid example.
- TN5000 dataset paper (2025) — newest open, biopsy-confirmed benchmark with baselines.
- ACR TI-RADS-based deep learning studies (e.g. 2021 multi-algorithm study reporting AUC ~0.90 on TR4/5) — directly your task.
- Multi-Attribute Attention Network for Interpretable Thyroid Nodule Diagnosis — attention + interpretability, closest in spirit to ThyroVision.
- A 2024–2025 review of deep learning in thyroid ultrasound CAD — find one recent survey and mine its reference list; it will hand you 40 more papers, organised.

8.3 Frontier (cite to show you know where it's going)

- MedSAM (Ma et al., 2024, Nature Communications); BiomedCLIP (Zhang et al., 2023); MedCLIP-SAM / v2 (2024–2025).
- LLNM-Net (2025, Nature Communications) — multimodal attention for thyroid.
- One ultrasound foundation-model paper (e.g. UltraSAM, 2024) to anchor the 'foundation models for ultrasound' trend.

The efficient move

Find ONE recent (2024–2025) survey specifically on 'deep learning for thyroid ultrasound' and ONE on 'explainable AI in medical imaging'. A good survey is a map of the whole subfield with a ready-made reference list and a 'future directions' section that literally lists open problems you can pick from. This is the fastest legitimate way to go from 0 to literate.

Part 9 — Open Problems (where your contribution can live)

Every one of these is a real, current bottleneck. Your project just needs to make a small dent in one or two.

- Generalisation / domain shift — models drop sharply on data from new machines/hospitals. Cross-dataset robustness is wide open.
- Small, imbalanced, weakly-labelled data — especially for under-represented populations (e.g. India). SSL/foundation models and clever augmentation are active answers.
- Label quality — many datasets use TI-RADS scores (a radiologist's opinion) rather than biopsy/pathology (ground truth). How much does this ceiling-cap models?
- Faithful explainability — do heatmaps actually reflect the model's reasoning, or just look reassuring? Quantifying trust is unsolved.
- Calibration — models are often over-confident; a '90% malignant' output must mean 90% for a clinician to use it for risk.
- Clinical integration & fairness — does the model work equally across age/sex/ethnicity? Does it actually reduce unnecessary biopsies in practice?

Translating an open problem into YOUR research question

Pick one and phrase it as a testable question, e.g.: 'Does adding spatial attention (CBAM) to a ResNet-50 improve benign/malignant sensitivity AND cross-dataset generalisation (train TN3K → test DDTI), and do the resulting Grad-CAM/attention maps localise to the nodule more faithfully than the baseline?' That one sentence is a complete, defensible intern project.

Part 10 — Practical Implementation Guide

Layer	Recommendation	Why
Language	Python	The entire ecosystem is Python. Non-negotiable
DL framework	PyTorch (with timm for pre-trained models)	Dominant in research; timm gives you ResNet/EfficientNet/ViT/ConvNeXt in one line
Imaging libs	OpenCV, Pillow, scikit-image; pydicom for DICOM; albumentations for augmentation	Standard preprocessing + augmentation stack
Classic ML	scikit-learn (metrics, SVM/RF baselines), XGBoost	Baselines + metric functions
Explainability	pytorch-grad-cam (Grad-CAM/++/Score-CAM), SHAP, Captum	Ready-made XAI
Segmentation (optional)	segmentation-models-pytorch, MONAI	MONAI is a medical-imaging-specific toolkit worth knowing
Hardware	Free GPUs: Google Colab / Kaggle Notebooks; then your lab's GPU	Don't buy anything. A ResNet on thyroid data trains fine on a free T4 GPU
Experiment tracking	Weights & Biases (free academic) or TensorBoard	Log every run — you WILL lose track otherwise
Code/version control	Git + GitHub	Reproducibility; mentors expect it

MONAI is worth a special mention

MONAI (Medical Open Network for AI, PyTorch-based) is built specifically for medical imaging — handles DICOM, medical augmentations, and has tutorials. Skimming its classification tutorial will teach you the correct medical-imaging workflow faster than generic PyTorch tutorials.

Part 11 — The ThyroVision Project Plan (bring this to your mentor)

A phased plan scoped to an internship running through mid-July. Each phase produces something showable, so even if you stop early you have a result.

Phase 0 — Foundation & framing (Week 1–2)

- Read Parts 0–2 of this doc + one thyroid-US-DL survey + one XAI survey. Build your paper log.
- Write a one-page problem statement in your own words and the single research question (Part 9 callout).
- Deliverable: that one-pager → first real mentor discussion.

Phase 1 — Data & baseline (Week 2–4)

- Get DDTI working end-to-end first (small, fast to iterate). Then bring in TN3K and/or TN5000.
- Build the pipeline: load → crop/clean artifacts → resize → normalise → patient-level split → augment.
- Train a fine-tuned ResNet-50 baseline. Report sensitivity/specificity/AUC with 5-fold CV.
- Deliverable: a working baseline with honest numbers + confusion matrix. This is already a real result.

Phase 2 — Attention (the core contribution) (Week 4–6)

- Add an attention module (CBAM or SE) to the same backbone. Keep everything else identical for a fair comparison.
- Compare baseline vs attention on the same splits. Did sensitivity/AUC improve? Is it statistically meaningful across folds?
- Deliverable: a clean baseline-vs-attention comparison table.

Phase 3 — Explainability & generalisation (Week 6–8)

- Generate Grad-CAM + attention maps. Show they localise to the nodule. Include both success and failure cases (honesty impresses).
- Run the cross-dataset test (train on TN3K → test on DDTI/TNUI). Does attention generalise better?
- Optional faithfulness check: mask the highlighted region and confirm performance drops.
- Deliverable: explainability figures + generalisation table. This is the heart of a paper.

Phase 4 (stretch) — Risk calibration & write-up (Week 8+)

- Calibrate probabilities; map to TI-RADS-aligned risk bands. Report calibration (reliability diagram / ECE).
- Draft a short paper / report: Intro, Related Work (your Part 8 log), Method, Experiments, Results, Limitations.
- Deliverable: a workshop-paper-shaped writeup, even if results are modest.

Scope discipline

If you're behind, ship Phases 1–3 well rather than all four badly. A rigorous baseline-vs-attention-with-explainability study on benign/malignant is a complete, honest contribution. Resist the urge to add modalities, multimodal text, or a custom transformer — that's how interns end up with nothing finished.

Part 12 — Roadmap: Beginner → Publication-Level

Stage	You can...	Milestone
Literate	Explain the problem, TI-RADS, and why it's hard, in plain English	Pass the 'explain it to your mentor' test
Operational	Run a pre-trained CNN on a dataset and read the metrics correctly	First working baseline
Comparative	Design a fair A/B experiment (baseline vs attention) and interpret it	A defensible comparison result
Rigorous	Do patient-level splits, cross-dataset tests, CV with error bars, explainability	Results a reviewer would trust
Contributing	Frame a gap, run the study, write it up with honest limitations	Workshop paper / thesis chapter
Independent	Read the frontier, critique it, and propose what's next	Lead your own follow-up

Most interns who finish strong reach 'Rigorous' and touch 'Contributing'. That is an excellent outcome and a real publication is genuinely achievable with a focused thyroid-attention study.

Part 13 — Questions to Ask Your Mentor (first meeting)

Walking in with sharp questions signals you've done the work. Ask these:

1. Scope: Is the goal benign-vs-malignant classification, TI-RADS category prediction, or full risk assessment? Which should I prioritise first?
2. Data: Do we have access to any private/clinical thyroid ultrasound data, or should I rely on public datasets (DDTI, TN3K, TN5000)? Is there any Indian dataset available?
3. Labels: For our data, are labels from TI-RADS scoring or biopsy-confirmed pathology? (This affects what the model can honestly claim.)
4. Novelty: Is the expected contribution a new method, a rigorous benchmark/comparison, or clinical validation? What would 'success' look like to you?
5. Compute: What GPU resources does the lab have, or should I plan around Colab/Kaggle?
6. Output: Are we targeting a conference/workshop paper, a thesis chapter, or an internal report? What's the timeline expectation?
7. Segmentation: Do you want a nodule-segmentation pre-step, or classify whole images directly?
8. Prior work: Is anyone in the lab already working on this, and is there existing code I should build on rather than start from scratch?

One thing to say out loud in that meeting

“My plan is to start with a fine-tuned ResNet baseline on a public thyroid dataset, add an attention module, evaluate with sensitivity/specificity/AUC plus a cross-dataset generalisation test, and use Grad-CAM/attention maps for explainability. Does that scope match what you had in mind, or would you steer it differently?” That sentence shows you understand the problem, the method, and the rigour — and invites correction.

Appendix — Glossary

Term	Meaning
AUC / ROC-AUC	Area under the ROC curve; threshold-independent measure of how well the model separates classes (0.5 = random, 1.0 = perfect).
Bethesda system	Standardised categories for thyroid biopsy (cytology) results, used after FNA.
CBAM	Convolutional Block Attention Module; adds channel + spatial attention to a CNN.
CNN	Convolutional Neural Network; the standard deep model for images.
Domain shift	Performance drop when test data differs from training data (e.g. new machine/hospital).
DICOM	The standard file format for medical imaging (pixels + metadata).
FNA	Fine-Needle Aspiration; needle biopsy of a nodule.
Grad-CAM	Gradient-weighted Class Activation Mapping; heatmap explainability method.
Nodule	A lump in the thyroid; mostly benign.
Segmentation	Outlining the exact pixels of a structure (vs classification = one label per image).
Sensitivity (Recall)	Fraction of true positives (cancers) correctly caught.
Specificity	Fraction of true negatives (benign) correctly cleared.
SSL	Self-Supervised Learning; pre-training on unlabelled data.
TI-RADS	Thyroid Imaging Reporting and Data System; clinical ultrasound risk-scoring scheme.
Transfer learning	Reusing a model pre-trained on other data, then fine-tuning on yours.
ViT	Vision Transformer; attention-based image model.

Sources underpinning this document include peer-reviewed literature and dataset papers on TI-RADS-based deep learning, thyroid ultrasound datasets (DDTI, TN3K, TN5000, TNSCUI), explainable and attention-based thyroid models, and 2024–2025 medical foundation-model work. Verify specific figures against primary sources before quoting them in formal writing.